INCH-POUND

MIL-STD-2104(OS)
7 March 1990
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WS 20642
25 October 1983

MILITARY STANDARD

WELDING, TUNGSTEN ARC, GAS, MANUAL (GTAW-MA) REQUIREMENTS FOR

AMSC A4915 AREA THJM

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FOREWORD

1. This military standard is approved for use by the Naval Sea Systems Command, Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

2. Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commanding Officer, Naval Ordnance Station, Standardization Branch (3730), Indian Head, MD 20640-5000, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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1. SCOPE

1.1 Scope. This standard covers the requirements for welding rocket motor case components by the manual gas tungsten arc welding (GTAW-MA) process.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

STANDARDS

MILITARY

MIL-STD-453 Inspection, Radiographic

MIL-STD-1595 Qualification of Aircraft, Missile and Aerospace Fusion Welders

MIL-STD-1949 Inspection, Magnetic Particle

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from: Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM E 8 Tension Testing of Metallic Materials (DoD adopted)

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

AMERICAN WELDING SOCIETY (AWS)

AWS A3.0 Welding Terms and Definitions (DoD adopted)

(Application for copies should be addressed to the American Welding Society, Inc., 550 NW LeJeune Road, P.O. Box 351040, Miami, FL 33135.)

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(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 **Order of precedence.** In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 General definitions. Unless otherwise defined herein, the definition of welding terms used in this standard are in accordance with AWS A3.0.

3.2 Incomplete penetration. Incomplete penetration is a lack of fusion in the root of the weld or gap left by a failure of the weld metal to fill the root. Incomplete penetration appears on the radiograph as a dark continuous or intermittent line in the middle of the weld (see figure 1).

3.3 Crack. A crack is a discontinuity produced by a fracture in the metal, and appears as a fine dark line, straight or wandering in direction (see figure 2).

3.4 Underfill. An underfill is a depression on the surface of the weld or root surface extending below the surface of the adjacent base metal (see figure 3).

3.5 Sharp weld edge. A sharp weld edge occurs at the edge of the weld bead and may be caused by either the grinding operation or the welding operation or both. A sharp weld edge has a sharp edge that does not blend in with the parent material. The sharp weld edge appears on the radiograph as a sharp indication but may not have a density change on the film (see figure 4).

3.6 Sharp-tail porosity. A sharp-tail porosity is any cavity having a sharp tail-like indication (see figure 5).

3.7 Porosity. Porosity is a cavity caused by entrapped gas (see figure 6).

3.8 Lack of fusion. Lack of fusion is a defect caused by lack of union between the weld metal and parent metal or both. Lack of fusion appears on the radiograph as a thin dark line with sharply defined edges. Depending upon the orientation of the defect with respect to the X-ray beam, the line may tend to be wavy and diffuse (see figure 7).

3.9 Tungsten inclusion. A tungsten inclusion is an inclusion of tungsten in the weld bead. The inclusion appears on the radiograph as a high density material with definite sharp boundaries and may have sharp edges (see figure 8).

3.10 Undercut. A groove melted into the base metal adjacent to the toe or root of a weld and left unfilled by weld metal (see figure 9).

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4. GENERAL REQUIREMENTS

4.1 Welding process. The manual gas tungsten arc welding process shall be in accordance with the requirements specified in this standard.

4.2 Equipment. Welding equipment, including torches, electrodes, and regulators, shall be capable of making welds that meet requirements specified herein when operated by a certified welder.

4.3 Materials.

4.3.1 Base material. The base material (including alloy types, alloy combinations, heat treat conditions, and material thickness) shall be as specified on the applicable drawing.

4.3.2 Welding materials.

4.3.2.1 Filler metal. Welding wire used as filler metal shall be capable of producing satisfactory welds when used by a certified welder and applicable welding equipment. Composition of the filler metal shall be suitable for producing weldments that meet requirements specified herein.

4.3.2.2 Welding electrodes. The welding electrodes shall be two percent thoriated tungsten.

4.3.3 Inert gases. Inert gases shall be of commercial quality and shall be not less than 99.9 percent pure. Vapor content of the gases shall be controlled to ensure that the dew point is not above minus 70 degrees Fahrenheit (-70°F).

4.3.3.1 Inert gas mixture. A mixture of argon and helium gases may be used where greater torch sensitivity or greater arc stabilization is required.

4.4 Welding procedures. All welding shall be accomplished using weld procedures as specified herein. A welding procedure shall be established for each weldment prior to welding the first production part. The welding procedure shall be certified as specified in 5.1.1 and shall be adhered to in all subsequent production welding of that part, except as provided in 5.3.5.

4.4.1 Cleaning. All fine scale, film, coatings, plating, carburization, nonfused, or nonbound materials, oil, grease, contamination, or materials that will cause difficulty in obtaining the desired vacuum or will contaminate the weld shall be removed prior to the welding. The surface shall be inspected to assure good metal-to-metal contact.

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4.4.2 **Welding.**

4.4.2.1 Welding log. A welding log shall be maintained for all welding performed on each component (see 5.2.3).

4.4.3 Preheat. Prior to welding, components shall be preheated to 400°F minimum. Preheat shall be maintained until welding is complete.

4.4.4 Post-weld thermal treatment. As soon as practical after welding, the welded assembly shall be heat treated in accordance with the applicable drawing requirements.

4.4.5 Workmanship. Welds shall be of complete penetration unless other-wise specified on the applicable drawing. Welds shall be clean and free from scale, dirt, foreign particle inclusion, porosity, undercutting, concavity, and discontinuities within the limits specified in table I.

4.4.6 Inspection. All welds shall be visually inspected and nondestructively inspected to ensure compliance with the requirements of this standard. The contractor shall ensure all inspection requirements have been met prior to submitting the welded items to the contracting activity for acceptance (see 5.3).

4.4.7 Repair welds. Unless otherwise specified in the welding procedures (see 4.4), defects in the completed weldments exceeding the limits of table I may be repaired. Repair welding shall be limited by demonstrated tests in accordance with the welding procedures.

4.4.7.1 Weld repair procedure. All repair welding shall be in accordance with the certified welding procedure (see 4.4).

4.4.7.2 Post-weld treatment. The repair welds and heat affected zones shall receive post-weld thermal treatments of 4.4.4.

4.4.7.3 Repair welding log. A repair welding log shall be maintained for all weld repairs (see 5.2.4).

4.4.7.4 Repair weld quality. Repaired weldments shall meet all applicable strength and test requirements specified herein and shall be inspected as specified in 5.3.

4.5 Manual welders. All manual welding shall be performed by welders certified as specified in 5.1.2.

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TABLE I. Allowable discontinuities in weldments.

Classification	Limits of Acceptability
Subsurface porosity (see 3.7)	a. Size and shape limits. The maximum dimension of a single void shall not exceed 25 percent of the thickness of the applicable size. Elongated porosity whose longest accumulative dimension is greater than five times the width of the smallest dimension is unacceptable. b. Adjacent porosity. Voids of 0.020 inch maximum dimension, and greater shall be separated by a minimum of three times the maximum dimension of the largest void. Voids that appear overlapping, touching, or connected when viewed normal to the weld surface shall be treated as a single void. There shall be no more than 10 voids in any 1 inch of weld regardless of size. c. Linear discontinuities. Three or more voids, regardless of size, grouped in a line are unacceptable if the line extends more than 0.25 inch and the voids occupy more than 50 percent of the line length.
Tungsten inclusions (see 3.9)	d. Size and shape limits. The maximum dimension shall not exceed 20 percent of the applicable weld size. Elongated inclusions shall meet the requirements of a. above. e. Spacing, distribution and quantity. Tungsten inclusions shall be included with, not in addition to, subsurface porosity. See b. and c. above.
Surface porosity	f. Size and shape limits. The maximum dimension of a single void shall not exceed 25 percent of the applicable weld size. Elongated porosity whose longest accumulative dimension is greater than three times the width of the smallest dimension is unacceptable.

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TABLE I. Allowable discontinuities in weldments - Continued.

Classification	Limits of Acceptability
	g. Adjacent porosity. Voids of 0.015 inch maximum dimensions and greater shall be separated by a minimum of three times the maximum dimension of the largest void. Voids that appear overlapping, touching, or connected shall be treated as a single void. There shall be no more than 5 voids in any 1-inch of weld regardless of size. h. Linear discontinuities. Linear discontinuities shall meet the requirements of c. above.
Other indications	Any of the following items shall be cause for rejection: 1. Incomplete penetration (see 3.2). 2. Cracks (see 3.3). 3. Underfill (see 3.4). 4. Sharp weld edge (see 3.5). 5. Sharp-tail porosity (see 3.6). 6. Lack of fusion (see 3.8). 7. Concavity that would reduce weld size or thickness below minimum requirements of applicable drawing. 8. Undercut (see 3.10).

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5. DETAILED REQUIREMENTS

5.1 Certification.

5.1.1 Welding procedure. Prior to welding the first production part, the welding procedure shall be certified as specified herein.

5.1.1.1 Qualification weld. A qualification weld shall be made in accordance with the welding procedure. The qualification weld shall simulate the production part with respect to section thickness, base material, heat treat condition, joint design and fitup. The weldment shall be radiographically inspected in accordance with MIL-STD-453 and shall meet limits of acceptability specified in table I.

5.1.1.1.1 Metallographic sections. The welded joint shall be sectioned transverse to the direction of welding to provide a minimum of three metallographic specimens. The specimens shall be polished, lightly etched and examined for conformance to specified requirements. The examination shall include as a minimum:

a. Overall joint fusion and root penetration.

b. Concavity, convexity and size of fillet.

c. Undercutting.

d. Inclusions and voids.

e. Cracks.

5.1.1.2 Tensile strength requirements. The contractor shall demonstrate a minimum 85 percent weld joint efficiency with respect to ultimate strength of the base material. A minimum of four each base material and welded tensile specimens shall be prepared and tested in accordance with ASTM E 8. Material thickness shall be 1/8 inch minimum. The specimens shall be subjected to all processing, including thermal treatments, prescribed for the actual production parts. Specimens shall be of rectangular cross section conforming to figure 6 or 7 of ASTM E 8. In welded specimens, the weld location shall be at the gage length center.

5.1.1.3 Welding procedure data. The welding procedure data shall be as specified in the contract or order (see 6.3).

5.1.2 Welders. All manual welders shall be qualified in accordance with the requirements of MIL-STD-1595 for the process, position, joint types and alloys to be welded.

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5.2 **Records.**

5.2.1 Welding procedures. The approved weld procedure data shall be maintained and made available to the welding operator each time parts or joints covered by the procedure are welded in production. Any weld procedure changed after initial approval shall require recertification approval in accordance with the requirements specified in 5.1.1.

5.2.2 Welders. The contractor shall maintain a list of all certified/ qualified welders and their assigned identification. A separate record of each welder's performance shall be maintained. Recertification/requalification of a welder shall be required as specified in MIL-STD-1595 or whenever the welder fails to meet the standard of workmanship specified in 4.4.5.

5.2.3 Welding logs. Welding logs shall be maintained as specified in 4.4.2.1. The log shall record and define the welding operations and shall include but not be limited to, the following:

a. Component name, part number and serial number.

b. Welding identification.

c. Filler wire composition.

d. Preheat temperature.

e. Postheat temperature.

f. Welding amperage and voltage.

g. Interpass temperature.

h. Repair record.

5.2.4 Repair welding logs. Repair welding logs shall be maintained as specified in 4.4.7.3. Information contained therein shall include, but not be limited to, the following:

a. Component name, part number and serial number.

b. Location of rework.

c. Cause of rework: cracking, porosity, or other defects.

d. Maximum length, width, and depth of each rework grindout.

e. Total number of repairs per weld.

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f. Preheat temperature.

g. Postheat temperature.

h. Length of time at temperature.

5.2.5 Rejection records. Records covering rejected welds and the corrective action taken shall be maintained by the contractor.

5.3 Inspection.

5.3.1 Materials.

5.3.1.1 Welding wire and electrodes. The welding wire and electrodes used in the process specified herein shall be inspected for conformance to the requirements of 4.3.2.

5.3.1.2 Inert gases. The inert gases and inert gas mixtures used in the process specified herein shall be inspected for conformance to the requirements of 4.3.3.

5.3.2 Visual examination. The exposed surfaces of all welded joints shall be visually examined for conformance to the requirements of 4.4.5.

5.3.3 Radiographic inspection. All welds, except as specified on the applicable drawing, shall be radiographically inspected in accordance with MIL-STD-453 for conformance to the requirements of 4.4.5. Whenever radiographic interpretation is in question with respect to root fusion in fillet welds, the contractor shall develop a radiographic standard for use as referee (see 6.4).

5.3.4 Magnetic particle inspection. All welds shall be inspected in accordance with MIL-STD-1949 for conformance to the requirements of 4.4.5.

5.3.5 Procedure review due to weld rejection. Failure of a production weld to meet quality requirements shall be cause for review of the established weld procedure. The cause of the unsatisfactory welding shall be determined. If investigation results in any change to the weld procedure, the changed weld procedure shall be recertified and approved in accordance with this standard. If the unsatisfactory welding is found to be the result of faulty equipment, defective materials, or noncompliance with the weld procedure, corrections shall be made as required.

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6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The manual gas tungsten arc welding process covered in this standard is intended for use in the manufacture of case components of the Standard Missile Mk 104 Dual Thrust Rocket Motor.

6.2 Issue of DODISS. When this standard is used in acquisition, the applicable issue of the DODISS must be cited in the solicitation (see 2.1.1 and 2.2).

6.3 Consideration of data requirements. The following data requirements should be considered when this standard is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference paragraph	DID Number	DID Title	Suggested Tailoring
Appendix, 10.1 and 5.1.1.3 of basic text	DI-MISC-80678	Certification/ Data Report	See appendix, 30.1

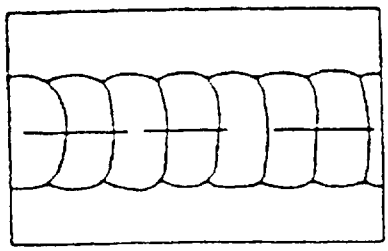
The above DID was cleared as of the date of this standard. The current issue of DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DID's are cited on the DD Form 1423.

6.4 Tailoring guidance. To ensure proper application of this standard, invitations for bid, requests for proposals, and contractual statements of work should tailor the requirements in sections 4 and 5 of this standard to exclude any unnecessary requirements. When considered necessary by the contracting activity, the contract should include provisions for Government approval of the contractor's radiographic standards for use as referee in questionable welds (see 5.3.3).

6.5 Subject term (key word) listing.

Rocket Motor, Dual Thrust, Mk 104
Standard Missile

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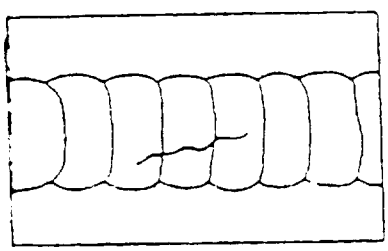


Radiographic projected appearance



Cross section of weld

FIGURE 1. Incomplete penetration.



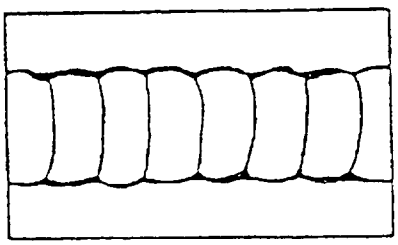
Radiographic projected appearance



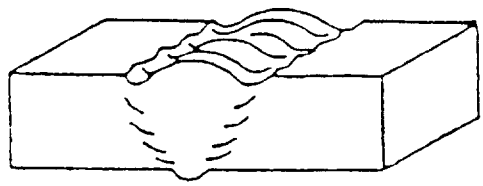
Cross section of weld

FIGURE 2. Cracks.

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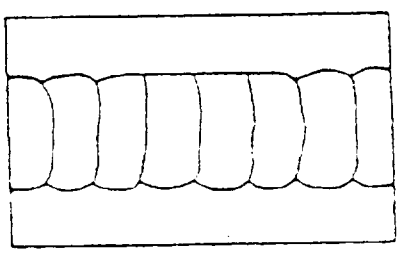


Radiographic appearance



Cross section
of weld

FIGURE 3. Underfill.



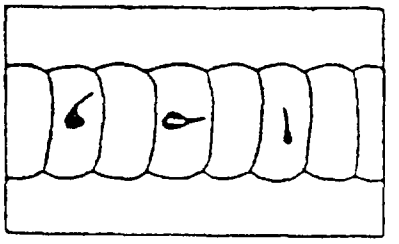
Radiographic appearance



May appear on top
or bottom bead

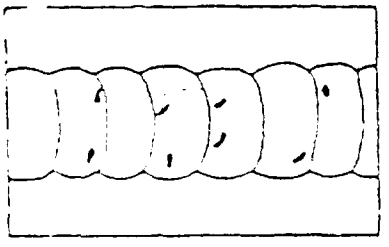
FIGURE 4. Sharp weld edge.

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Radiographic appearance

FIGURE 5. Sharp-tail porosity.



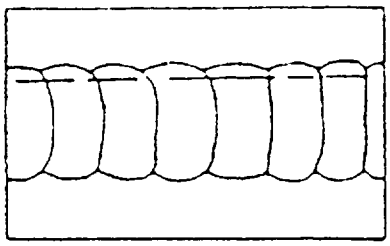
Radiographic projected appearance



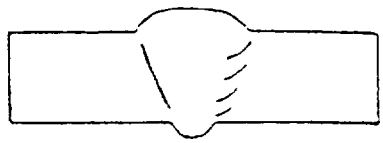
Cross section of weld

FIGURE 6. Porosity.

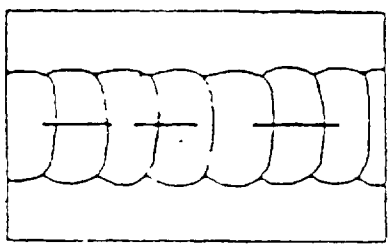
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Radiographic appearance



Lack of fusion at edge of weld

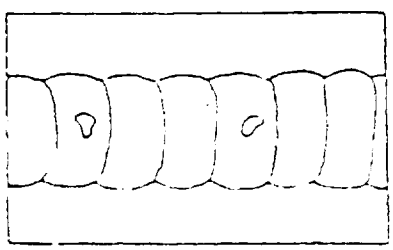


Radiographic appearance



Lack of fusion at center of weld. Note complete penetration.

FIGURE 7. Lack of fusion.



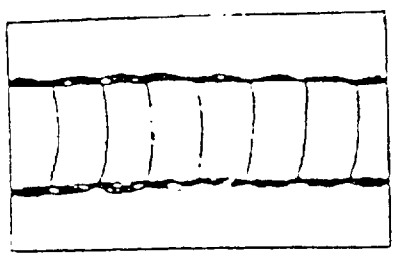
Radiographic appearance



Cross section of weld

FIGURE 8. Tungsten inclusion.

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Radiographic appearance

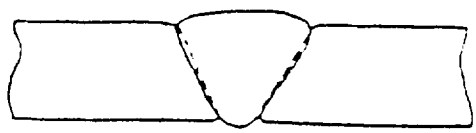


FIGURE 9. Undercut.

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APPENDIX

CERTIFICATION DATA/REPORT

TECHNICAL CONTENT REQUIREMENTS

10. SCOPE

10.1 Scope. This appendix details the technical content of certification data/report covering the contractor's manual gas tungsten arc welding process control for the welding of rocket motor case components. This appendix is mandatory only when data item description DI-MISC-80678 is cited in the contract or order (see 6.3).

20. APPLICABLE DOCUMENTS

This section is not applicable to this appendix.

30. TECHNICAL CONTENT REQUIREMENTS

30.1 Welding procedure data. The welding procedure data for each joint shall include, as a minimum, the following information as applicable to the welding procedure to be certified as specified in 5.1.1 of this standard:

a. Cleaning procedures.

b. Preheat and postheat: temperature, time and method.

c. Weld wire.

d. Welding amperage and voltage.

e. Welding equipment.

f. Tooling and fixtures.

g. Number of weld passes.

h. Weld repair (including demonstrated tests for).

Preparing activity:
Navy - OS
(Project THJM-N276)

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2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-STD-2104(OS)	2. DOCUMENT DATE (YYMMDD) 900307

3. DOCUMENT TITLE
WELDING, TUNGSTEN ARC, GAS, MANUAL (GTAW-MA) REQUIREMENTS FOR

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

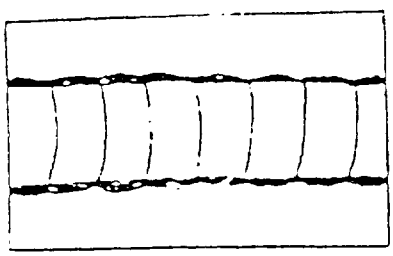
a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME COMMANDING OFFICER NAVAL ORDNANCE STATION	b. TELEPHONE (Include Area Code) (1) Commercial (301)743-4510	(2) AUTOVON 364-4510
c. ADDRESS (Include Zip Code) ATTN: CODE 3730 INDIAN HEAD, MD 20640-5000	[illegible]	

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Radiographic appearance

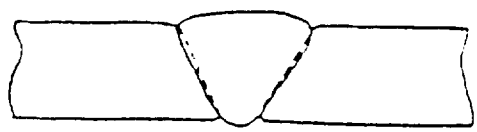


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ATTN: CODE 3730 INDIAN HEAD, MD 20640-5000	[illegible]	

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